DETR-based Predictive Railway Adhesion Management

Mathematical Formulation

1 Problem Definition

Consider a railway locomotive equipped with:

- a forward-facing camera providing original RGB images;
- a BEG module providing an overlay representing track geometry;
- an existing system measuring wheel slip and providing traction-state signals;
- an active normal-force augmentation actuator controlled through a force-command interface.

The objective is to estimate, predictively, the additional normal force required to maintain an adequate adhesion margin under the observed visual and dynamic conditions.

Let t denote the current time and k the index of a longitudinal window in the visual scene. The input set is

$$\mathcal{X}_t = \{I_t,\ M_t,\ s_t,\ v_t,\ \tau_t,\ \gamma_t,\ h_{t-1}\},$$

where:

Symbol	Definition
I_t	Original RGB image, $I_t \in \mathbb{R}^{H\times W\times 3}$
M_t	BEG track-geometry overlay, e.g. $M_t \in [0,1]^{H\times W}$
s_t	Measured wheel slip; scalar or per-axle vector
v_t	Locomotive speed, in m/s
τ_t	Requested traction torque, in N m, or a normalized representation
γ_t	Track gradient, represented as a grade fraction or angle under a defined convention
h_{t-1}	Previous latent temporal state

The dynamic inputs are optional for a minimal vision-only realization, but are recommended to distinguish predicted visual risk from the locomotive's actual intervention demand.

2 Visual Fusion and Feature Extraction

Define two visual encoders:

$$E_I = \phi_I(I_t;\theta_I), \qquad E_M = \phi_M(M_t;\theta_M),$$

where ϕ_I extracts features from the original image and ϕ_M encodes the geometry supplied by BEG.

The representations are spatially aligned and combined:

$$Z_t^{(0)} = \mathrm{Fuse}(E_I, E_M).$$

The fused representation is projected into a latent space of dimension d:

$$Z_t^{(1)} = W_f Z_t^{(0)} + b_f + P_{\mathrm{pos}},$$

where P_{pos} denotes positional encoding.

The BEG overlay defines *where* the track should be analysed, while the RGB image provides information about *how* the surface appears. A new semantic-segmentation network is not required by this proposed architecture.

3 DETR Encoder–Decoder with Specialised Queries

The Transformer encoder models global relationships between scene regions:

$$Z_t = \text{TransformerEncoder}(Z_t^{(1)}).$$

Define N_q learnable queries:

$$Q^{(0)} = \{q_1^{(0)}, q_2^{(0)}, \ldots, q_{N_q}^{(0)}\}, \qquad q_i^{(0)} \in \mathbb{R}^d.$$

The decoder updates the queries through self-attention and cross-attention over the scene representation:

$$Q_t^{(\ell)} = \text{DecoderLayer}\left(Q_t^{(\ell-1)}, Z_t\right), \qquad \ell = 1, \ldots, L,$$

with final decoder output

$$Q_t = Q_t^{(L)}.$$

Instead of queries dedicated exclusively to object detection, the proposed architecture organises query groups around railway-condition estimation:

$$Q_t = \left\{Q_t^{\text{texture}}, Q_t^{\text{continuity}}, Q_t^{\text{surface}}, Q_t^{\text{rail}}, Q_t^{\text{environment}}, Q_t^{\text{adhesion}}\right\}.$$

These groups may contain one or more queries each, depending on the desired granularity.

3.1 Track-Texture Head

Define the texture representation

$$\mathbf{z}_{\text{texture},t} = f_{\text{texture}}(Q_t^{\text{texture}}).$$

It represents multiscale texture, regularity, and local surface structure. Its conceptual components may be written as

$$\mathbf{z}_{\text{texture},t} = \left[z_{\text{fine}}, z_{\text{coarse}}, z_{\text{periodic}}, z_{\text{irregular}}, z_{\text{contrast}}\right].$$

These components are learned latent features, not direct physical measurements.

3.2 Surface-Continuity and Compatibility Heads

Define a continuity score:

$$\hat{c}_t = \sigma\left(f_{\text{continuity}}(Q_t^{\text{continuity}})\right),$$

and a distribution over surface classes:

$$\hat{\mathbf{p}}_{\text{surface},t} = \text{softmax}\left(f_{\text{surface}}(Q_t^{\text{surface}})\right).$$

Possible conceptual classes include conventional ballast and sleepers, continuous concrete, level crossings, bridges or open track structures, and unknown or insufficiently observable surfaces.

An actuator-compatibility score is defined as

$$\hat{a}_{\text{surface},t} = f_{\text{compat}}\left(\mathbf{z}_{\text{texture},t}, \hat{c}_t, \hat{\mathbf{p}}_{\text{surface},t}\right).$$

This score is a visual estimate of compatibility, not a direct measurement of negative pressure or airflow.

3.3 Rail-Condition and Environmental Heads

The rail-condition distribution is

$$\hat{\mathbf{p}}_{\text{rail},t} = \text{softmax}\left(f_{\text{rail}}(Q_t^{\text{rail}})\right),$$

and the environmental-context distribution is

$$\hat{\mathbf{p}}_{\text{environment},t} = \text{softmax}\left(f_{\text{environment}}(Q_t^{\text{environment}})\right).$$

These heads may estimate visible leaves, apparent wetness, visible contamination, and seasonal context. They contribute to adhesion-risk estimation but do not directly measure the physical coefficient of friction.

4 Temporal Model

To exploit scene evolution and locomotive dynamics, introduce a recurrent latent state:

$$h_t = \text{GRU}\left(h_{t-1}, [\text{Pool}(Q_t); s_t; v_t; \tau_t; \gamma_t]\right),$$

where semicolons denote vector concatenation and Pool aggregates query representations.

A GRU is one possible implementation of temporal state; other temporal aggregation mechanisms may be substituted.

For predictive operation, visual information should be associated with the track segment that will be traversed, rather than only the locomotive's current position. A future-condition head may estimate:

$$\hat{\mathbf{c}}_{t+T_p} = f_{\text{future}}(h_t),$$

where $\hat{\mathbf{c}}_{t+T_p}$ denotes estimated conditions for a future segment at prediction horizon T_p.

5 Adhesion-Demand Model

The adhesion head estimates a wheel-slip risk score:

$$\hat{r}_t = \sigma\left(f_{\text{risk}}\left(h_t, \mathbf{z}_{\text{texture},t}, \hat{\mathbf{p}}_{\text{rail},t}, \hat{\mathbf{p}}_{\text{surface},t}\right)\right), \qquad \hat{r}_t \in [0,1].$$

This score represents risk, not the coefficient of friction or wheel slip itself.

The measured wheel slip may be compared with a control target $s^\star$:

$$e_t = \max\left(0,\ |s_t - s^\star| - \delta\right),$$

where δ is an allowed tolerance. For multiple axles, define

$$\mathbf{e}_t = [e_{1,t}, e_{2,t}, \ldots, e_{n,t}].$$

An adhesion-demand head may then compute

$$\hat{D}_t = f_D\left(h_t, \hat{r}_t, \mathbf{e}_t, v_t, \tau_t, \gamma_t\right),$$

where $\hat{D}_t$ represents the estimated intervention demand in a latent or physical representation selected for the model.

An optional effective-adhesion estimate is

$$\hat{\mu}_t = f_\mu(h_t, \mathbf{e}_t, \tau_t, v_t).$$

This estimate is conditional on available dynamic observations. Where the available observations do not identify adhesion reliably, uncertainty should be retained rather than assigning an unsupported precise value.

6 Physical Reference Model for Normal-Force Augmentation

For a reference locomotive with mass

$$m_{\text{ref}} = 84{,}000 \text{ kg},$$

the approximate static weight is

$$N_{\text{ref}} = m_{\text{ref}} g$$

and hence

$$N_{\text{ref}} = 84{,}000 \cdot 9.81 \approx 824.0 \text{ kN}.$$

This is the approximate static weight of the locomotive, not automatically the normal load on driven axles or the required additional force.

Under a Coulomb-type approximation, the maximum longitudinal force transferable through wheel–rail contact is

$$F_{\text{adh,max}} \approx \mu N_{\text{drive}},$$

where N_{drive} is the relevant normal load on the driven axles.

If the requested traction force is $F_{\text{traction,req}}$, an idealised estimate of the required additional normal load is

$$\Delta N_{\text{ideal}} = \max\left(0,\ \frac{F_{\text{traction,req}}}{\mu} - N_{\text{drive}}\right).$$

This relationship illustrates the physical principle but is not sufficient by itself to command the actuator. Axle-load distribution, vehicle dynamics, and effective adhesion limits must also be considered.

7 Final Force-Request Head

Define a joint regression vector:

$$u_t = \big[h_t;\ \mathbf{z}_{\text{texture},t};\ \hat{c}_t;\ \hat{a}_{\text{surface},t};\ \hat{\mathbf{p}}_{\text{rail},t};\ \hat{\mathbf{p}}_{\text{environment},t};\ \hat{r}_t;\ s_t;\ v_t;\ \tau_t;\ \gamma_t\big].$$

The neural regression head computes

$$\boxed{\hat{F}_{\text{raw},t} = f_F(u_t; \theta_F)}$$

where $\hat{F}_{\text{raw},t}$ is the raw requested normal-force augmentation in kN.

To constrain the output to be non-negative and bounded, one possible parameterisation is

$$\hat{F}_{\text{raw},t} = F_{\max}\, \sigma\left(f_F(u_t; \theta_F)\right),$$

where $F_{\max}$ is a design limit for the actuator interface. Alternatively, a softplus activation may be followed by saturation.

The command sent to the actuator-control system is

$$\boxed{F_{\text{cmd},t} = a_t\ \text{sat}_{[0,F_{\max}]}\left(\hat{F}_{\text{raw},t}\right)}$$

where $a_t \in \{0, 1\}$ is a command-validity indicator and sat limits the command to the allowed interval.

The validity indicator may be determined by a separate validation mechanism based on uncertainty, surface compatibility, actuator state, and safety limits.

For a complete output interface, define

$$\boxed{\mathcal{Y}_t = \left[F_{\text{cmd},t}\ [\text{kN}],\ \hat{\sigma}_{F,t},\ a_t\right],}$$

where $\hat{\sigma}_{F,t}$ represents estimated uncertainty in the force prediction.

The neural network requests a force; the actuator controller determines the internal commands needed to realise it, subject to actuator capability and constraints.

8 Training Objective

The architecture is multi-task, estimating visual conditions, risk, and force demand. A possible total loss is

$$\begin{aligned}\mathcal{L}_{\text{total}} = \lambda_{\text{tex}}\mathcal{L}_{\text{tex}} + \lambda_{\text{surface}}\mathcal{L}_{\text{surface}} \\ + \lambda_{\text{rail}}\mathcal{L}_{\text{rail}} + \lambda_{\text{env}}\mathcal{L}_{\text{env}} \\ + \lambda_{\text{risk}}\mathcal{L}_{\text{risk}} + \lambda_F\mathcal{L}_F + \lambda_{\text{temp}}\mathcal{L}_{\text{temp}},\end{aligned}$$

where $\lambda_i \geq 0$ are task weights.

8.1 Classification Loss

For classification heads, cross-entropy may be used:

$$\mathcal{L}_{\text{class}} = -\sum_{j=1}^{C} y_j \log \hat{p}_j.$$

This may supervise surface classes, rail conditions, and environmental context when corresponding labels are available.

8.2 Force-Regression Loss

For a reference force demand $F_{\text{ref},t}$, one possible loss is

$$\mathcal{L}_F = \text{Huber}\left(\hat{F}_{\text{raw},t} - F_{\text{ref},t}\right).$$

8.3 Temporal Loss

To penalise unjustified changes in force demand:

$$\mathcal{L}_{\text{temp}} = \sum_t \left|\hat{F}_{\text{raw},t} - \hat{F}_{\text{raw},t-1}\right|.$$

This penalty should be conditioned on actual changes in risk and locomotive state; it should not prevent rapid force-command increases when a wheel-slip hazard emerges.

These equations define a possible training objective. They do not imply that training data or measured reference-force labels already exist. A real implementation would require physically justified supervision and separate safety validation.

9 Compact End-to-End Formulation

The complete conceptual model can be summarised as

$$\boxed{\begin{aligned} Z_t &= \text{Encoder}(I_t, M_t), \\ Q_t &= \text{DETRDecoder}(Z_t, Q^{(0)}), \\ \mathbf{c}_t &= f_{\text{condition}}(Q_t), \\ h_t &= f_{\text{temporal}}(h_{t-1}, Q_t, s_t, v_t, \tau_t, \gamma_t), \\ \hat{F}_{\text{raw},t} &= f_F(h_t, \mathbf{c}_t), \\ F_{\text{cmd},t} &= a_t \ \text{sat}_{[0,F_{\max}]}(\hat{F}_{\text{raw},t}). \end{aligned}}$$

Here $\mathbf{c}_t$ includes estimated features for track texture, surface continuity, rail condition, and environmental context.

The conceptual contribution is the transformation of visual track information and locomotive dynamic state into a predictive normal-force augmentation request, expressed in kN and transmitted through an interface independent of the actuator's internal implementation.